

Dr. TENDAI MASUNDA ZENGENI

(PhD, MSc, BSc)



My name is Dr. Tendai Masunda Zengeni and I have extensive experience as a researcher in the field of Optical Communications, telecommunications and data communications with 11 SCOPUS International Journal publications as well as serving as a Reviewer in 10 SCOPUS indexed journals and other local journals. With a PhD in Fiber Optics and Nanophotonics, which falls under applied physics, obtained from Universiti Utara Malaysia and one year of post-doctoral research in High-Speed Optical Computing from University of West Attica, Athens, Greece, I have a wealth of knowledge and experience in this field and I am passionate about my work and have dedicated many years to developing my expertise in this area. Currently, I am affiliated with Africa University, Mutare, Zimbabwe as a Chairperson at the College of Engineering & Applied Sciences. I was appointed as Board Vice Chairperson for Netone cellular, Zimbabwe and I am also a UNESCO African Women in Tech & AI Fellow 2024, trained at the International Center for Artificial Intelligence – Morocco. I am also a member of the UNESCO Women for Ethical AI (W4EAI) which focuses on AI Capacity building for girls and women. I was also appointed as a UNESCO AI Ethics Expert without Borders and a Core Team member drafting the Zimbabwe AI Strategy. In October 2025 I was appointed Regional Director for Southern Africa at Women In STEM (Wi-STEM) Africa and in April 2026 appointed an Advisor to the Zimbabwe National Innovation and Acceleration Centre.

DETAILS

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EMPLOYMENT HISTORY

Year	Position	Institution	Duties
October 2025 to present	Regional Director Southern Africa (pro bono)	WISTEM AFRICA (Women in STEM Africa)	• Lead the strategic development and operational execution of the organization's mission across the Southern Africa region. • Spearheading program launches, partnership development, and community engagement to empower women and girls in STEM. • Managing a regional team and budget to deliver on key objectives and increase organizational impact. • Forging strategic partnerships with corporations, academic institutions, and NGOs to secure support and drive initiatives.

Year	Position	Institution	Duties
August 2025 to present	Chairperson	Africa University, College of Engineering and Applied Sciences. Department of Artificial Intelligence, Software Engineering & Computer Science.	Appointed H.O.D of the Department of Artificial Intelligence, Software Engineering & Computer Science College of Engineering and Applied Sciences. Administrative position in the governance of the University. The Head of Department is at a central point in the administration, representing the College providing leadership in the development of innovative programmes aligned with the vision and mission of the University as well as managing the affairs of the Department.
June 2025	UNESCO AI Ethics Expert (pro bono)	UNESCO AI Ethics Experts Without Borders	https://www.unesco.org/ethics-ai/en/ai-ethics-experts-without-borders/tendai-zengeni?hub=47 UNESCO's AI Ethics Experts Without Borders (AIEB) network is a global initiative comprising of leading experts from over 50 countries, providing policy guidance, legislative advice, and capacity building tailored to national needs. The network is funded by the European Union and the Patrick J. McGovern Foundation.
January 2025	UNESCO Women 4 Ethical AI (pro bono)	UNESCO Women 4 Ethical AI (W4EAI) Expert	UNESCO, through its Recommendation on the Ethics of Artificial Intelligence, aims to provide a basis to make AI systems work for the good of humanity, individuals, societies, the environment, and ecosystems, and to prevent harm- ensure an ethical AI lifecycle. To accelerate this goal, the organization launched Women4Ethical AI (W4EAI) a platform that unites leading female AI experts at different stages and phases of life, from diverse backgrounds.
July 2023 to Present	Deputy Chairperson of the Board of Directors	Netone Cellular Private Limited, Zimbabwe	Oversight and Strategic Guidance to drive the vision of the Netone (Pvt) Limited business. Upholding and curating the strategic and corporate governance tenets of the business. Assist and advise the Chairman on establishing agendas for board meetings. Acting as an additional key point of contact with the CEO and Management in the Chair's absence or incapacity, and perform the responsibilities of the Chairman when the Chairman is unavailable.
January 2022 to Present	Lecturer	College of Engineering and Applied Sciences Africa University, Mutare, Zimbabwe	Lecturing BSc Computer Science (Theory of Computing, Simulation and Modeling, Advanced Networks and Protocols, Computer Architecture and Design, Project Management, Research Methodology, Software Project Management, Data Communications and Computer Networks, Technical Communication).

Year	Position	Institution	Duties
May 2023 to July 2023	Part time Lecturer	Manicaland State University of Applied Sciences, Mutare, Zimbabwe	Lecturing BSc Computer Science (Cyber Law and Professional Ethics and Psychology of Computing, Human Computer Interaction).
June 2015 to May 2023 (8 yrs)	Senior Research Fellow	Optical Computing Research Laboratory affiliated with the Department of Electrical Engineering & Computer Science, Massachusetts Institute of Technology (MIT), Cambridge, USA.	Conducting high speed optical research on • Multimode fiber • Single Mode fiber • Photonic Crystal fiber • Tapered Fiber • Few Mode Fiber • Multi-core Fiber
March 2018 to October 2019 (8mths)	Research Assistant FRGS GRANT	Optical Computing Research Laboratory, Universiti Utara Malaysia	Code:S/O13571 Mode-Wavelength Division Multiplexing for Separating Optical Network Units into Virtual Point to Point Connections for Smart Services in Internet of Things: Increasing Data Rates with less Optical Fiber
June to Dec 2013 (6mths)	Intern	School of International Studies, (SOIS) Universiti Utara Malaysia, Malaysia	Web designer. Creating website for SOIS
Jan 2005 to Dec 2007 (2yrs)	Bank Clerk & Admin Assistant	Diplomatic Centre, Reserve Bank of Zimbabwe, Harare, Zimbabwe	Diplomatic Centre Bank Transactions, Administration duties and Organizing Monetary Policy Presentation by the Governor for the Diplomats
Jan 2000 to Dec 2003 (3yrs)	Computer Technician /Computer Trainer (ICDL)	Accurate Services, Bulawayo, Zimbabwe	Lecturing Microsoft packages, Servicing computers, networking, computer repairs and support.
Jan 2002 to Dec 2012 (10yrs)			EXPERIENCE IN THE ARTS, BROADCAST INDUSTRY MALAYSIA • Directed a low budget musical video as a project for Bachelor of Media Technology, Video directing class, composed the song, translated it to Chinese and Malay languages recorded the song, edited it, shot the video and edited the video. • Produced a PSA for domestic violence under UniFilm Malaysia, Universiti Utara Malaysia

			• Trained in Single Camera and MultiCamera production for live broadcast • Trained in Radio Broadcast, as a Producer and also a Radio Dj • Trained in Movie Colour Grading • Participated in shooting an advertisement and was the lighting assistant as well as the Clapperboard talent
			EXPERIENCE IN THE ARTS, BROADCAST INDUSTRY ZIMBABWE • Performing Artist Live at Amakhosi theatre, Bulawayo • Performing Artist at Miss Malaika Beauty Pageant 2002, which was broadcast worldwide • Performing Artist at Muzee Bira Galas, Independence Galas, Unity Gala, Heroes Gala and other musical Galas organized by the Ministry of Information • Recording Artist of the Sisonke musical Project and shooting the videos across the monuments of Zimbabwe that is Victoria Falls, Boterekwa, Matopos Hills, Kopje Harare, Kariba Dam • Recording Artist in Score Warriors National Project to inspire the Zimbabwe Warriors with various artists • Composed a PSA song, recorded it and shot a video with the International Organization of Migration for an AIDS awareness Campaign for travelers • Wrote scripts for many voice-over projects and also provided the voice talent

ACADEMIC QUALIFICATIONS

Year	Position	Institution	Duties
July 2010 to July 2014	Bachelor of Science (Hons) Information Technology with a major in Networking (2.1) CGPA 3.16/4	Universiti Utara Malaysia (UUM)	Courses including but not limited to Operating System, IT Project Management, Advanced Programming, Software Engineering, Artificial Intelligence, Networking Basics, Network & System Security, Computer System Architecture and Network Management.
July 2013 to July 2014	Bachelor of Media Technology	Universiti Utara Malaysia (UUM)	Tv & Radio Broadcasting Courses taken and appearing on transcript • Online Radio Production • Video Directing • Digital Broadcasting Management • Multimedia System Networking Skills Taught • Radio Production • Leadership Skills • Script Writing • Multi Camera Tv Production • Single Camera Tv Production • Video Directing • Radio Production • Radio DJ • Floor Management • Recording audio and Mixing • Lighting and Clapperboard
July 2014 to July 2015	Master of Science in Information Technology - (Fiber Optics) (First Class) CGPA3.65/4	Universiti Utara Malaysia (UUM)	Information Technology with specialization in (Fiber Optics) Title of MSc Project paper: FFE & DFE Equalization in WDM-MDM over MMF Specializing in Networking and Fiber Optical Communications, Multimode Fiber, Mode Division Multiplexing, and Equalization.
Sept 2016 to May 2023	PhD in Fiber Optics/ Nanophotonics	Universiti Utara Malaysia (UUM)	Thesis Title: Simulation of Elliptical Core Few Mode Fiber Switch in Mode Division Multiplexing for Enhancing Bandwidth Performance. Area of specialization: Multimode Fiber, Few mode Fiber, Mode Division Multiplexing, Photonic Crystal Fiber, Digital Signal Processing, Equalization, Image Processing, Optical Encryption (cybersecurity and artificial intelligence), Data transmission.

Year	Position	Institution	Duties
March 2022 to March 2023	ERASMUS Post-Doctoral Fellowship	University of West Attica (UNIWA), Athens, Greece	European Union (EU) Sponsored Post-Doctoral research on high speed optical computing and artificial intelligence at the Department of Electrical and Electronics Engineering, Faculty of Engineering, University of West Attica, Athens, Greece. Published a paper titled: Large Core Germanium Doped Photonic Crystal Fiber over FMF
APRIL 2024 TO July 2024	African Women In Tech & AI Fellow (AWITAI)	- The International Center for Artificial Intelligence Center Morocco (Ai Movement) - University Mohamed VI Polytechnic (UM6P) - UNESCO - OCP Foundation	This program supports African women with an interest in technological innovation and Artificial Intelligence, implemented in the framework of the Flagship program 4 of UNESCO's Operational Strategy for Priority Africa 2022-2029 "Harnessing new and emerging technologies for sustainable development in Africa". The 16 courses taught were: - Course 1 – Welcome Remarks, Women's Leadership Hard Skills, Intro to AI - Course 2 - AI in Africa: Survey of successful Implementations of AI - Course 3 – Startup Innovation - Course 4 - Introduction to Python and Google Colab - Course 5 - AI Project Workflow/ Collecting, cleaning and Storing the Data - Course 6 - AI Project Workflow/ Collecting, cleaning and Storing the Data - Course 7 - Developing and Training ML models: Data Driven Models - Course 8 - Developing and Training ML models: Data Driven Models - Course 9 - Developing and Training ML models: Data Driven Models - Course 10 - Developing and Training ML models: Reinforcement Learning - Course 11 -Decision Support – Constraint Programming - Course 12 - Decision Support – Decision Aid - Course 13 - Introduction to MLOps - Course 14 - Prompt Engineering - Course 15 - Ethical Issues in AI – Data Governance - Course 16 – Regulations regarding AI – Transformations by Ai (ChatGPT) The modules covered were comprehensive in teaching all aspects of Artificial Intelligence.

PROFESSIONAL QUALIFICATIONS

Year	Position	Institution	Duties
2026	International Elections Observer	- African Union - Ministry of Foreign Affairs, Kingdom of Morocco	Short term Election Observer
2014	Android Mobile Application Development	Computing Enrichment and Development Division (COPED)	Universiti Utara Malaysia (UUM) Learning how to develop mobile apps using Android Studio
2006	I.C.D.L – (International Computer Driving License)	In-house training, Reserve Bank of Zimbabwe	MODULE 1,2,3,4,5,6,7 Mod 1: Computer Essentials Mod 2: Online Essentials Mod 3: Word Processing Mod 4: Spreadsheets Mod 5: Database Mod 6: Slide Presentation Mod 7: IT Security
2002	Higher National Diploma in Micro-Computer Technology Part 1, 2 & 3	Computer Networking (City & Guilds) University of Zimbabwe - Department of Electrical Engineering	The Computer Technician Syllabus included: Fundamental concepts of microprocessors, microcontrollers, computer memory organization assembly language programming, computer architecture, and interfacing, embedded systems, control systems, along with practical applications and troubleshooting and maintenance.

AWARDS AND RECOGNITION

Year	Position	Institution	Duties
April 2026	Short Term Elections Observer	- African Union - Ministry of Foreign Affairs, Kingdom of Morocco	Appointed a short term elections observer to oversee elections where deployed, on the African Continent
April 2026	Advisor	POTRAZ National Innovation and Acceleration Centre	Appointed into the Advisory for The Centre. Overseeing the national innovation ecosystem and supporting local commercialization of home bred digital innovations.

Year	Position	Institution	Duties
October 2025	Executive Board Member	WiSTEM Africa (Women in STEM Africa)	Strategic oversight of the development and operational execution of the organization's mission across the Southern Africa region. Spearheading partnership development and community engagement to empower women and girls in STEM regionally.
August 2025	Appointed Expert in the Core Team for drafting the NATIONAL AI STRATEGY FOR ZIMBABWE	UNESCO Ministry of ICT, Postal and Courier Services (MICTPCS)	One of the Core team made up of 4 experts, being the only woman among them, drafting the AI Strategy for Zimbabwe.
August 2025	Appointed Chairperson	College of Engineering & Applied Sciences, Africa University, Mutare, Zimbabwe	The Head of Department is at a central point in the administration, representing the College providing leadership in the development of innovative programmes aligned with the vision and mission of the University as well as managing the affairs of the Department.

June 2025	Co-opted as a member	UNESCO AI Ethics Experts without Borders	The AIEB network comprises leading experts from over 50 countries, providing policy guidance, legislative advice, and capacity building tailored to national needs.
Jan 2025	Appointed Chairperson of the National Mobile BroadBand (NMBB) Committee	NetOne Cellular Private Limited, Zimbabwe	Curating the National Mobile Broadband infrastructure at NetOne Cellular. Making oversight decisions and advising the management when need be.
Jan 2025	Appointed in the Government Thinktank for Zimbabwe AI Readiness Assessment (RAM)	UNESCO Ministry of ICT, Postal and Courier Services (MICTPCS)	Expert consultation bringing together selected experts in AI, ICT, and Digital Transformation, from government, academia and to write the RAM report for Zimbabwe to be included in the United Nations Report. https://unesdoc.unesco.org/ark:/48223/pf0000394685 I was representing Africa University
Jan 2025	Inducted as MEMBER	UNESCO's Women 4 Ethical AI (W4EAI) Expert Platform	A platform that unites leading female AI experts at different stages and phases of life, from diverse backgrounds.
June 2024	Appointment to the Solar Construction Project Committee,	Africa University, Mutare, Zimbabwe	To oversee the implementation of specific solar installation project(s); Review project progress and provide guidance and recommendations to ensure that project goals and outcomes are met timeously.

Year	Position	Institution	Duties
April 2024 to October 2025	Appointed Advisor in CompED 2025 Regional Program Advisory Committee	ACM Botswana and Australia	ACM Global Computing Education Conference, Gaborone Botswana
April 2024 to July 2024	African Women in Tech & AI AWITAI Fellow – Awarded UNESCO Scholarship	- The International Center for Artificial Intelligence Center Morocco (Ai Movement) - University Mohamed VI Polytechnic (UM6P) - UNESCO - OCP Foundation	African Women in Information Technology & Artificial Intelligence Program
13 Feb 2024	Appointed Chairperson of the Integrity Committee Commissioned by Zimbabwe Anti-Corruption Commission	NetOne Cellular Private Limited, Zimbabwe	Handling reports on ethical conduct of the employees and the business is addressing the issues around the conduct of employees and eradication of corrupt activities and adherence to processes and procedures to ensure that the internal control environment remains sound.

Feb 2024	Appointed Judge: iShub CHANS pitch competition	Africa University, Mutare, Zimbabwe	This competition is a pivotal platform for the College of Health, Agriculture and Natural Sciences students to showcase their groundbreaking ideas, with the ultimate aim of driving innovation and entrepreneurship in health, agriculture and natural sciences. The winning team will be awarded up to \$3,000 in seed capital to bring their innovative solutions to life.
Feb 2024	Appointed Honorary Peer Reviewer	Council of Peer Reviewers Global Journals Massachusetts Incorporated, USA	Global Journal of Computer Science and Technology Global Journals serves as an international platform dedicated to facilitating the publication of high-quality research and review papers across all fields of study. It aims to connect researchers, inventors, scientists, engineers, managers, doctors, and professors engaged in various research disciplines, fostering a comprehensive environment for academic and professional growth.

Year	Position	Institution	Duties
Jan 2024	Appointment into The College Promotions Committee	Africa University, Mutare, Zimbabwe	Appointed by the Deputy Vice Chancellor, Africa University to be part of the College Promotions Committee for the College of Engineering & Applied Sciences to consider an application for promotion from Senior Lecturer to Associate Professor.
July 2023	Appointed Vice Chairperson for The Board Of Directors	NetOne Cellular Private Limited, Zimbabwe	Assisting the Chairman in preparing meeting agendas and other materials also helping to facilitate discussions and ensure smooth meeting operations. Taking on specific assignments or projects as delegated by the Chair. Like all board members the Vice Chair, has a fiduciary duty to the company and its shareholders. Participating in board meetings, offering input on strategic decisions, and monitoring the company's performance.
July 2023	Appointed Board Member	NetOne Cellular Private Limited, Zimbabwe	Non-Executive Director with an oversight role for the business, ensuring corporate governance
March 2022 to March 2023	Post-Doctoral Research Fellow	ERASMUS Mobility Scholarship, University of West Attica, Athens Greece	Post-Doctoral research on high speed optical computing and artificial intelligence at the Department of Electrical and Electronics Engineering, Faculty of Engineering, University of West Attica, Athens, Greece.

2018	BEST PAPER	IEEE International Conference on Photonics (ICP) 2018	Title: Modal Properties of a Varied High Indexed Large Core 4 Mode Photonic Crystal Fiber. IEEE & Photonics Society Malaysian Chapter
June 2015 to Dec 2018	Awarded UUM POSTGRADUATE SCHOLARSHIP	Universiti Utara Malaysia (UUM)	Awarded the Postgraduate Scholarship by the OFFICE OF THE DEPUTY VICE CHANCELLOR, RESEARCH & INNOVATION 2015/2018 (Session A151)
Jun 2015 to Dec 2020	Senior Research Fellow	Optical Computing Research Laboratory	University Utara Malaysia Research Area: Fiber Optics

July 2011	Appointed iGreen Africa Goodwill Ambassador	iGreen Africa, Zimbabwe	Forming international partnerships with international bodies, institutions and persons in line with Millennium Development Goals (MDGs) 7&8. Thinking, Talking and Acting green to bring a long- lasting solution and legacy that will transcend beyond generations to follow.
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REVIEWER EXPERIENCE

Appointed as Peer Reviewer for SCOPUS academic publications for these International Journals:

	Database	Position	Journal
1.	SCOPUS	Reviewer	Journal of Optical Communications (De Gruyter)
2.	SCOPUS	Reviewer	Journal of Engineering Science and Technology (JESTEC)
3.	SCOPUS	Reviewer	Advances in Mathematics: Scientific Journal
4.	SCOPUS	Reviewer	Advances in Science, Technology & Engineering Systems
5.	SCOPUS	Reviewer	(JENRS) Journal of Engineering Research and Sciences
6.	SCOPUS	Reviewer	(Optics) Science Publishing Group
7.	SCOPUS	Reviewer	ICECET'22 IEEE
8.	SCOPUS	Reviewer	ITU Kaleidoscope Academic Conferences
9.		Reviewer	Global Journal of Computer Science and Technology
10.		Reviewer	Journal of Science and Engineering for Sustainable Development
11.	SCOPUS	Reviewer	Zimbabwe Conference on Information and Communication Technology (ZCICT)
12.	SCOPUS	Reviewer	ACM Global Computing Education Conference 2025
13.	AUREC	Reviewer	Research Ethics Committee Africa University
14.	Mutare Poly	Reviewer/ Editor	Mutare Polytechnic and Africa University joint Research Conference
15.	POTRAZ	Reviewer	POTRAZ Research Journal (Zimbabwe)

CURRICULUM DEVELOPMENT

	University	Position	Program
1.	Africa University, College of Engineering and Applied Sciences	Thought Leader	Bachelor of Science Honors in Cyber Security June – December 2024
2.	Africa University, College of Engineering and Applied Sciences	Thought Leader	Master of Science in Artificial Intelligence For August – December 2026

MEMORANDUM OF AGREEMENT

	University	Position	Program
1.	Memorandum of Agreement (MOU) between Africa University and Netone Cellular Signed October 2024	Facilitator	Facilitated the MOU for Academia-Industry collaboration between Africa University and NetOne Cellular where a mutually beneficial relationship was forged.
2.	MOU between Africa University and ZESA or ZETDC	Facilitator	Status – Discussions Ongoing MOU to be Finalized by midyear 2025
3.	MOU between Africa University and University of West Attica, Athens, Greece	Facilitator	Status – Activated 1st Cohort to go in January 2026 MOU to be finalized by 1 st Quarter 2026

CONFERENCES AND WORKSHOPS

Year	Name	Venue	Details
2018	BEST PAPER IEEE International Conference on Photonics (ICP)	Langkawi, Malaysia	Winner Best Paper Title: Modal properties of a varied high indexed large core 4-mode photonic crystal fiber
April 2024	Adjudicator	POTRAZ 2024 ICT Research Competition	Judging innovations at the POTRAZ 2024 ICT Research Competition
13 May, 2024	Director of Ceremonies	Africa University, College of Engineering & Applied Sciences In conjunction with the Office of the Deputy Vice Chancellor	Public Lecture by Associate Professor Dr. Kristin Kuter Fullbright Scholar, St. Mary's College, Notre Dame, Indiana , USA Topic: Data Science as a Career in Today's Technology Ecosystem
June 24 to 28, 2024	Ecosystem Stakeholder/ Advisory	The Universal Service Fund (USF) Innovation Drive Postal & Telecommunications Regulatory Authority of Zimbabwe (POTRAZ)	The National Acceleration Centre Design Workshop was held in Kadoma, Zimbabwe. This integral event was used to lay the fundamentals for the establishment of the National Innovation Acceleration Centre. This workshop was strictly by invite and was focused on Engaging with key stakeholders from different groups such as finance, academia, the private sector, the public sector, entrepreneurs from within the country to provide expertise crucial to the strategic and operational validation of the Centre's design. The objective was to align the Centre with national digital transformation goals and global best practices, enhancing its capability to drive innovation across various sectors
19 July 2024	AI /DIGITAL CONFERENCE	Harare, Zimbabwe	Hosted by Zimbabwe Newspapers Group in conjunction with POTRAZ and NetOne Cellular Embracing cutting edge technology for the 4th Industrial Revolution. Bringing together experts and stakeholders to explore the transformative impact of digital technologies on various sectors. Running under the theme 'Navigating the AI Revolution: Data, Pricing, and Taxation in Zimbabwe', the conference aimed to foster collaboration and knowledge sharing, highlighting the significance of adopting AI
30 July, 2024	Presenter Capacity Building Workshop	Africa University, Mutare	Topic: Collaboration in and out of the classroom - Establishing clear guidelines for behavior and collaboration - Promoting collaborative learning - Utilizing collaborative tools in and outside the classroom

Year	Name	Venue	Details
28 August 2024	Panelist - AI Ethics and Governance Symposium	hosted by IndabaX AI Zimbabwe	Platform to engage in discussions that will steer the Zimbabwean assimilation of Artificial Intelligence in the Global community
Sept 2024	Panelist and Judge	POTRAZ ICT INNOVATION EXPO	Running under the Theme: "Building heritage and community driven solutions" Young scientists showcased their inventions and innovations to solve real problems that we face in Zimbabwe and the world
Sept 2024	Facilitator AI Capacity Building	Media Institute of Southern Africa (MISA) ZIMBABWE Volunteering	I engaged practitioners from various facets of the media fraternity (print, online, radio, tv) having a conversation on emerging technologies and how they affect our media in Zimbabwe. It was organized by MISA Zimbabwe. In attendance, to name a few, were representatives from ZBC, Manica Post, freelance journalists, Wasu Post, the Ministry of Information, Publicity and Broadcasting Services, a Newspaper Cartoon illustrator, and Diamond FM. I emphasized that as a nation we need to familiarize ourselves, demystify and harness the power of AI in all aspects of industry, social life, economy and governance because AI is here to stay
October 2024	Presenter	Launch of POTRAZ ICT Research Journal	POTRAZ Journal launch and Symposium gathering key thought leaders and industry stakeholders to share ideas and foster collaborative discussions around ICT developments in Zimbabwe
October 2024	Mentor	UBUNTUNET Alliance Women Hackathon 2024 Dar es Salaam, Tanzania	Participated in the Hackathon and came out NUMBER 1 out of 7 countries across Africa. The hackathon, aimed to foster innovation among young women across Eastern and Southern Africa, with a specific focus on research and education My Team presented a project to bridge the digital divide using feature phones and USSD to provide educational material for underserved communities
Nov 2024	Facilitator AI Capacity Building	Development literacy Institute, Zimbabwe Volunteering	In-person training on Technology, Economic Development and Law. Training attendees on emerging technologies and on artificial intelligence, block chain, renewable energy, 5G technology, focusing on their transformative potential for economic growth in Zimbabwe. Incorporating case studies highlighting technology driven success stories and risks in developing countries to illustrate the practical impact of emerging technologies

Year	Name	Venue	Details
Nov 2024	Panelist	Computer Society of Zimbabwe's Ignite, Innovate, Thrive Summit 2024	Experts discussing how to navigate digital solutions for a sustainable tomorrow. This is also the Society's 50th Anniversary Summit
Nov 2024	ZIMCHE AI Framework Drafting Group	National AI Framework for Universities	Drafting the National AI Framework for universities in Zimbabwe as a team of 12 Academics sanctioned by the Zimbabwe Council for Higher Education (ZIMCHE)
Nov 2024	Round Table Panelist	AfricaCom Tech Festival, Capetown	Uniting global technology and telecom leaders, entrepreneurs, and policymakers to drive digital transformation and innovation across Africa. I gave my insights on AI and emerging technologies in Zimbabwe.
Nov 2024	Panelist	Digital Health Stakeholders Indaba	The Digital Health Stakeholders Indaba serves as a platform for dialogue and collaboration, bringing together government representatives, donors, implementing partners, and industry experts to explore strategies and innovations to advance Zimbabwe's digital health transformation. Discussion on Building the Technological Backbone for Digital Health in Zimbabwe
Feb 2025	Expert – Think Tank	National AI Ethics and Readiness Assessment Report UNESCO	Deliberating with 24 other national experts on the readiness of Zimbabwe in terms of AI adoption
March 2025	Judge or Adjudicator	World Robot Olympiad Zimbabwe Chapter	Judging during the hackathon and contributing my knowledge and expertise until we crowned the winner
March 2025	Mentor	ISACA IT Bowl, Puerto Rico	Guiding my team during the hackathon and going over the challenge. My team was number 3 countrywide
March 2025	Mentor	Africa University, i5Hub Hackathon	Guiding my team during the hackathon and going over the challenge presented
March 2025	Judge and Adjudicator	POTRAZ 2024 ICT Research Competition	Judging during the POTRAZ 2024 ICT Research Competition. Using my expertise in reviewing research proposal which were submitted for the Postal and Telecommunications Regulatory Authority of Zimbabwe (POTRAZ) 2024 Information and Communication Technology (ICT) Research Competition. POTRAZ hosted the ICT Research Competition for state-owned Institutions of Higher Learning (IHL) in Zimbabwe, under the theme "Creating Citizen-Centric Smart and Sustainable Solutions

Year	Name	Venue	Details
March 2025	Expert – Thinktank	AI RAM ZIMBABWE VALIDATION WORKSHOP UNESCO	One of the experts validating the UNESCO AI Readiness Assessment Methodology that was deliberated in February 2025 where the final document will be presented before the UN Secretary General António Guterres in Paris, France
May 2025	Domain Expert	2025 POTRAZ Innovation Drive Ideathon, Masvingo	Hackathon showcasing Ideas from Innovators with prize monies amounting to USD\$8000 to 15 winners. I was a judge as well as a mentor
May 2025	Innovation Ecosystem Builder	POTRAZ Strategic Foresight Co-Creation Workshop, Gweru	POTRAZ, Universal Services Fund, Innovation Drive, ITU for the strategic planning of the National Acceleration Centre
May 2025	Expert	U4SSC Thematic group on Digital Public Infrastructure for Cities	U4SSC Secretariat Telecommunication Standardization Bureau International Telecommunication Union
June 2025	Speaker	Zimbabwe Institute of Strategic Thinking (ZIST)	Stakeholder's Inaugural Conference. Unlocking Public Sector Efficiency in the context of Vision 2030 – Strategy as an Accelerator. My presentation was on Unlocking ICT Efficiency in the Context of Vision 2030.
June 2025	Speaker	UNESCO Global Forum on the Ethics of Artificial Intelligence, Bangkok, Thailand	Speaking at the Women Leaders at the Forefront of AI Policy and Technology session, powered by UNESCO Women4Ethical AI Platform. Shared my insights on Ethics of AI in Africa juxtaposed with the global village as well as my experience in terms of emerging technologies and its adoption in developing countries.
July 2025	Speaker	Global Business Achievers Network	Speaking on the Networking and Awards Night about the impact of AI and emerging technologies on business.
July 2025	Panel Speaker and Adjudicator	ITU (International Telecommunications Union) POTRAZ 2025 Innovation Expo and Conference, Bulawayo	Hosted by the Postal and Telecommunications Regulatory Authority of Zimbabwe (POTRAZ) and ITU. This inaugural event will be instrumental in accelerating the digital economy through building localized capacity, global partnerships and leveraging open technologies for sustainable development.
August 2025	Panelist	The Vase The Women and Leadership Summit 2025	SESSION 5 PANEL DISCUSSION: “HARNESSING TECHNOLOGY: WOMEN PIONEERS IN AI AND DIGITAL TRANSFORMATION” Panel discussion featured women leaders who are at the forefront of technological innovation. The discussion focused on how they leverage AI and emerging technologies to drive change and growth within their organizations, with interactive discussions on overcoming barriers to women's participation in tech leadership.

Year	Name	Venue	Details
Sept 2025	Speaker	Zimbabwe Institute of Strategic Thinking	Priming of Strategic Institutions towards the delivery of NDS2- attainment of Vision 2030 targets. I presented submissions from the Artificial Intelligence Desk.
Sept 2025	Panelist	- African Women Leaders Network (AWLN) - African Union - UN Women	Seventh Intergenerational Retreat- Session on AI and Emerging Technologies AI Leadership Dialogue: Knowledge Exchange for AI. I shared my knowledge on the ethical considerations of AI, also gave a brief account of my experience drafting the National AI strategy for Zimbabwe as well as inspiring upcoming scientists on my AI journey as an expert in the field.
Sept 2025	Panelist	UNESCO Regional Office for Southern Africa's Artificial Intelligence for Information Accessibility (AI4IA) Conference	My contribution to the conversation was on AI and the Youth: Shaping the Next Generation of Innovators and Ethical Leaders in Zimbabwe. I focused on the proactive role young innovators must play in the AI ecosystem.
October 2025	Speaker	AI Education Conference, Manicaland Province, Mutare, Zimbabwe	Presented on future trends in AI and Education to Manicaland province educators from various high schools and universities and polytechnics. This was to capacitate educators and familiarize them with the importance of AI in this digital age.
October 2025	Adjudicator	Africa University, Mutare, Zimbabwe	Judging the Cybersecurity competition to commemorate the Cybersecurity awareness month. The competition was between students from Africa University and Manicaland State University of Applied Sciences.
October 2025	Session Chair	POTRAZ ICT Research Symposium, Masvingo	Chairing 2 sessions at the Research Symposium and providing guidance to manuscript authors and preparing them for the final publication of their research. First session was E Waste: A Problem or an Opportunity Perception of Urbanites from Masvingo City Ward 6. The second session was Challenges of Implementing Cybersecurity and Data Protection Measures in Business.
October 2025	Domain Expert	POTRAZ and ITU Strategic Foresight Co-Creation Workshop Kariba	Being prepared for leadership of the Zimbabwe National Innovation and Acceleration Centre
Nov 2025	Participant	Africa Tech Festival Cape town South Africa	Attending the prestigious festival showcasing high tech, modern trends and latest technology in the AI era around telecommunications and data communications.

Nov 2025	Speaker	Zimbabwe National Defense University	International Peace and Security Conference (ICPS 2025). Theme: AI as a catalyst for Peace, Security and Development.
Feb 2026	Speaker	Black History Month Africa University	High level panel discussion on African Futures and the afterlives of Black History
Feb 2026	Speaker	Gap Solutions AI Education Conference	Topic presented: AI Tools for smarter teaching
March 2026	Speaker/ Facilitator	Parliament of Zimbabwe Capacitating the Parliamentary Portfolio Committee on ICT about AI	Topic presented: Unpacking the pillars of the Zimbabwe National AI Strategy to the Parliamentarians
March 2026	Speaker and Panelist	Women in Science Innovation and Technology Conference 2026	Theme: Advancing the contribution of Women In Science, Technology And Innovation Towards Research Innovation And Entrepreneurship Towards Vision 2030
March 2026	Speaker and Panelist	Academic Integrity and assessments in an ai world conference, Johannesburg, South Africa	Topic presented: Balancing Innovation With Integrity In Academic Assessments In Africa
April 2026	Speaker and Panelist	POTRAZ Innovation Product Launch and business Forum	Innovation Drive and the Launch of the NIAC Topic presented: Unpacking the Zimbabwe AI Strategy
April 2026	Speaker and Panelist	3 rd Dialogue on Elections and democracy in Africa	Organized by the African Union Department of Political Affairs, Peace and Security together with the Kingdom of Morocco' Ministry of foreign affairs Topic presented: AI, the Youth and Democracy

SUPERVISION

No.	Project Title	Description
1.	Supervising 11 Undergraduate Student Dissertations	Supervising students from BSc Artificial Intelligence, Software Engineering, Computer Science and Computer and Information Systems. All Graduated in June 2025
2.	Supervising 15 Undergraduate Student Dissertations	Supervising students from BSc Artificial Intelligence, Software Engineering, Computer Science and Computer and Information Systems. To Graduate in June 2026

AI BOARD TRAINING

No.	BOARD Trained	Description
1.	ZINWA BOARD	October 2025. Trained the ZINWA Board of Directors and Senior Management on how to leverage AI for Strategic Leadership and Empowering the ZINWA Board for the Future
2.	ZIMSWITCH BOARD	October 2025. Trained the ZIMSWITCH Board of Directors and Senior Management on how to leverage AI for Strategic Leadership and Empowering the Board for the Future in the AI era.

RESEARCH ACTIVITIES:

No.	Project Title	Description
1.	Increasing throughput for Multimode Fiber in MDM Systems	Increasing the carrying capacity of multimode fiber MMF, using mode division multiplexing (MDM) in the broadband network. Modal dispersion, being the major bandwidth limiting factor in MMF can be mitigated through selective modal excitation.
2.	Evaluating performance of MDM schemes	Evaluation of the digital signal processing and propagation level of the system using metrics namely the Bit Error Rate (BER), eye diagrams, spatial analyzer, power coupling coefficient, mean square error, differential mode group delay.
3.	Designing new components to accommodate spatial capabilities	DQPSK, WDM, MDM, design in order to include the spatial laser capabilities because it had the normal laser for single mode transmission, therefore I had to configure and program the component (VCSEL), modifying it to have spatial capabilities in Optisim.
4.	Using Artificial intelligence for MDM	Artificial intelligence is a compelling subject of today. Algorithms can be used in order to select the best modes either by classification or clustering. Artificial neural networks, or support vector machines, genetic algorithms using Matlab can be harnessed in order to extract the perfect modes in the fiber hence optimizing transmission. We tell the light what to do.
5.	Equalization	During data transmission a lot of signal interference occurs (noise) in the channel and through mode coupling in MMF. Equalization attempts to regenerate the signal as well as clean the data, and make the data arrive at the same time. There are many techniques towards equalization as well as using artificial intelligence algorithms.
6.	Photonic Crystal fiber	Currently I am working on novel techniques to design new structures of PCF to contribute knowledge in research.
7.	Data Center Switching	Data Centers continue to grow and therefore there is a need to manage the switching in the fiber interconnects as they become more and more complicated. My research seeks new switching architecture so as to future proof bandwidth and switching demand within the data centers.

SKILLS & PROFICIENCIES:

No.	Title	Description
1.	MMF (multimode fiber)	Digital signal processing mainly equalization using AI and hardware. Simulations using Optsim, Beamprop as well as Matlab.
2.	WDM (wavelength division multiplexing)	Optimization of the channels by multiplexing using Wavelength Division Multiplexing in order to increase data capacity per channel.
3.	MDM (mode division multiplexing)	Optimization of the channels by multiplexing using Mode Division Multiplexing techniques in order to further increase data channels in multimode fiber.
4.	OFDM	Orthogonal Frequency Division Multiplexing being one of the many multiplexing schemes used to complement and optimize data channels to increase capacity.
5.	Matlab, Optsim, Beamprop	Simulation software proficiency when designing topology as well as simulations to determine the performance of components and data transmission.
6.	Mode coupling	Inter Symbol Interference and Mode Coupling Effects on data transmission and ways to mitigate them using Artificial Intelligence for Equalization.
7.	Sensors	Using multimode fiber as sensing technology e.g. underwater oil pipes
8.	Photonic Crystal Fiber	Design of photonic crystal fiber for optimization in multimode fiber data transmission

PUBLICATIONS:

CURRENT SCOPUS INDEX = 4 (search Tendai Masunda)

1. Masunda, T., & Amphawan, A. (2018). Feedforward equalizers for mdm-wdm in multimode fiber interconnects. *Journal of Optical Communications*, 39(2), 147-153. **SCOPUS** <https://ui.adsabs.harvard.edu/abs/2018JOC...39..147M/abstract>
2. Masunda, T., Amphawan, A., & Al-dawoodi, A. (2017). Effect of decision feedback equalizer taps on 3×6-channel mode-wavelength division multiplexing system performance in multimode fiber. In *EPJ Web of Conferences* (Vol. 162, p. 01012). EDP Sciences. **SCOPUS** https://www.epj-conferences.org/articles/epjconf/abs/2017/31/epjconf_incape2017_01012/epjconf_incape2017_01012.html
3. Masunda, T., Amphawan, A., Alshwani, S., & Aldawoodi, A. (2018, April). Modal properties of a varied high indexed large core 4-mode photonic crystal fiber. In *2018 IEEE 7th International Conference on Photonics (ICP)* (pp. 1-3). IEEE. **SCOPUS** <https://ieeexplore.ieee.org/document/8533166>
4. Fareed, A., Masunda, T., Ghazi, A., Fakhrudeen, A. M., Aljunid, S. A., Idrus, S. Z. S., & Amphawan, A. (2019). Comparison of different wavelength propagations over few-mode fiber based on space division multiplexing in conjunction with electrical equalization. *International Journal of Electronics and Telecommunications*, 65(1), 5-10. **SCOPUS** <https://ijet.pl/index.php/ijet/article/view/10.24425-123558>
5. Amphawan, A., & Masunda, T. (2020, January). Mode generation in elliptical core few mode fiber. In *AIP Conference Proceedings* (Vol. 2203, No. 1, p. 020008). AIP Publishing LLC. **SCOPUS** <https://aip.scitation.org/doi/abs/10.1063/1.5142100>
6. Ghazi, A., Fareed, A., Masunda, T., Aljunid, S. A., Idrus, S. Z. S., Amphawan, A., & Fadhel, I. E. I. (2020, April). Spiral-Phased Laguerre-Gaussian Modes Generation in SWDM over Few Mode Fiber based on Electrical Equalization. In *Journal of Physics: Conference Series* (Vol. 1529, No. 2, p. 022012). IOP Publishing. **SCOPUS** <https://iopscience.iop.org/article/10.1088/1742-6596/1529/2/022012>
7. Abdulbaqi, A. S., Najim, S. A. M., Mohammed, F. M., Masunda, T., & Panessai, I. Y. (2021). Wireless EEG Transmission and Evaluation Based On iCloud Efficiency: Age of Telemedicine. *Journal of Engineering Science and Technology*, 16(3), 2416-2425. **SCOPUS** https://jestec.taylors.edu.my/Vol%2016%20issue%203%20June%202021/16_3_40.pdf
8. Mahadzir, N. A., Amphawan, A., Masunda, T., Menon, P. S., Jalar, A., Bakar, M. A., & Ismail, A. G. (2021). Tapered waveguide design for mode conversion in mode division multiplexing (MDM). *Electromagnetics*, 41(6), 448-458. **SCOPUS** <https://www.tandfonline.com/doi/full/10.1080/02726343.2021.1986783>
9. Zengeni, T. M., Angeli, C., Din, R., & Amphawan, A. Large Core Germanium Doped Photonic Crystal Fiber Over FMF. 8th International E- conference on Engineering, Technology and Management - ICETM 2022 <https://theired.org/conference/paper/large-core-germanium-doped-photonic-crystal-fiber-over-fmf-4156>

10. Awasthi, Y., Garikayi, T., Zengeni, T. M., & Makambwa, T. (2024). A Framework for Cloud Migration in Academic Institutions. International Journal of Open Information Technologies, 12(7), 88-96. <https://cyberleninka.ru/article/n/a-framework-for-cloud-migration-in-academic-institutions/viewer>
11. Tendai Masunda Zengeni, Roshidi Bin Din, Angela Amphawan, Paul Simbarashe Mupfiga Garikayi Talon (2024). Phase Reconstruction of the LG Mode Using Photonic Crystal Fibre and the Gerchberg-Saxton Algorithm for Cyber Security, POTRAZ ICT Research Journal <https://www.potraz.gov.zw/wp-content/uploads/2024/10/Phase-Reconstructuion-of-the-LG-Mode-using-Photonic-Crystal-Fiber-and-the-Gerchberg-Saxton-Algorithm-fro-Cyber-Security.pdf>
12. Optical Asymmetric Public Key Cryptography Based On Modified Gerchberg-Saxton Algorithm Book Chapter - Pending Publication
13. Zimbabwe National AI Strategy <https://zimaistrategy.net/>

RESEARCH GRANT AWARD AND PARTICIPATION

Grant name: FRGS, Institution: Universiti Utara Malaysia
Research Center: Optical Computing & Technology Research Laboratory
Project Title: Digital Signal Processing for Mode-Wavelength Division Multiplexing Start
Date: 2016-09-11
End Date: 2018-08-11
Funding Agency: Ministry of Higher Education (MOHE)
Project Code from Awarding Agency: FRGS/1/2016/STG02/UUM/02/1

Description/Background:

Internet-of-Things allows devices to sense other devices and interact and communicate with them. Sensing devices from smart services generate an enormous volume of data traffic. However, due to their small sizes, sensing devices in Internet-of-Things have limitations in terms of memory, computing power, accessibility. Thus, storage, processing and networking capabilities are gathered from remote computers.

The propagation time for data acquisition, retrieval and integration of datasets from one sensing device to another in Internet-of-Things is critical and may be limited by the amount of data channels available. In order to improve the propagation time, multiplexing schemes for enhanced connectivity from a single optical fiber to various devices are utilized. Although wavelength division multiplexing (WDM) is conventionally used in multimode optical fiber backbones in local area networks, it will not be sufficient to support the enormous bandwidth required by Internet-of-Things. Mode-wavelength division multiplexing (MWDM) is a promising solution to increase the bandwidth provided by an optical fiber backbone. This project focuses on digital signal processing in MWDM in a real optical experiment.

REFERENCES:

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